AIOps Aeronautical Engine Model

HoloLens Data Collection Guide for First-Time Collectors

This guide turns the ordered reference photos into a clean, step-by-step collection script. Collectors should speak the message, perform the action, and pause briefly so the HoloLens recording clearly captures the final state.

Image 1 (1.jpeg)

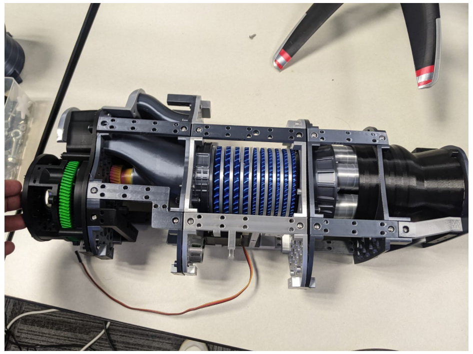

Image 16 (16.jpeg)

Image 20 (20.jpeg)

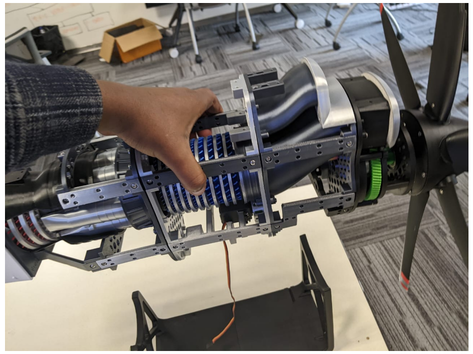

Image 35 (35.jpeg)

Key rule	Propeller rule	Recording rule
Most connections require 2, 4, or 6 screws. The action is not complete until all required screws for that interface are started and seated.	Use the small screws staged in ordered Image 30 for the propeller hub. Do not mix them with larger frame or cowling screws.	Keep the part, screw, screwdriver tip, and both hands visible. If the view is blocked, repeat the action slowly.

Before Recording

1. Put all parts on a clean table and keep the screw area visible.
2. Start HoloLens recording before touching the model.
3. Say the step message before each action.
4. Use one action per step and pause after completion.
5. If a mistake happens, keep recording and recover in the same take.
6. At the end, hold the final assembly in view for at least five seconds.

Assembly Collection Steps

Each step below shows its reference photo(s) at full size, in ordered sequence. Image numbers are ordered sequence numbers; the source filename is shown in parentheses.

STEP 01 Step 1 of 24

Lay out the engine model parts

Image 1 (1.jpeg)

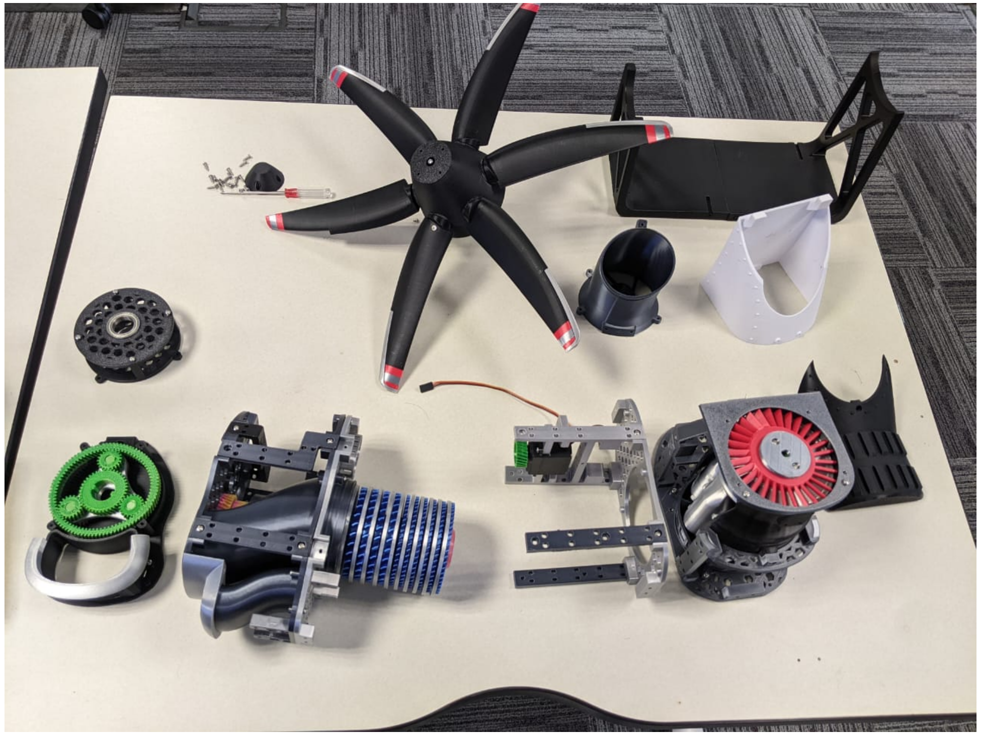

Image 2 (2.jpeg)

Say: Start with the engine model parts laid out.

Do: Show the propeller, rear hub, cylindrical core, frame rails, fan/intake module, cowling, black covers, screwdriver, and fasteners.

Visible check: Hold the full layout steady before assembly begins.

STEP 02 Step 2 of 24

Prepare the rear green-ring hub

Image 3 (3.jpeg)

Say: Pick up the rear green-ring hub module.

Do: Handle the rear hub module and rotate it so the connector interface is visible.

Visible check: The green ring and mounting face should be clear in view.

Secure the rear hub circular plate

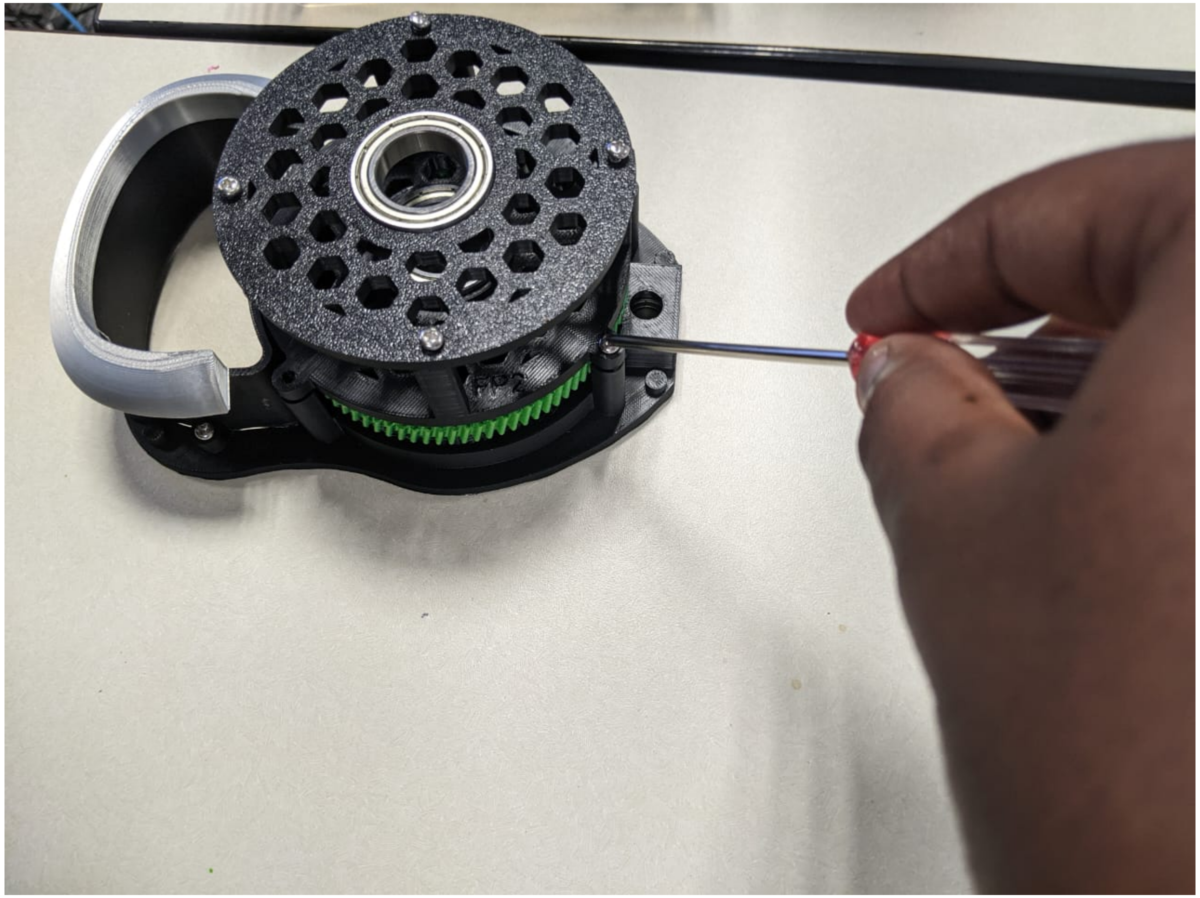

Image 4 (4.jpeg)

Say: Connect the rear hub using the circular fastener pattern.

Do: Use the screwdriver to fasten the circular rear hub or bearing plate.

Visible check: Use the required 4 or 6 screws for this circular interface. Capture the screwdriver touching the screw head.

STEP 04 Step 4 of 24

Position the cylindrical core on the bracket

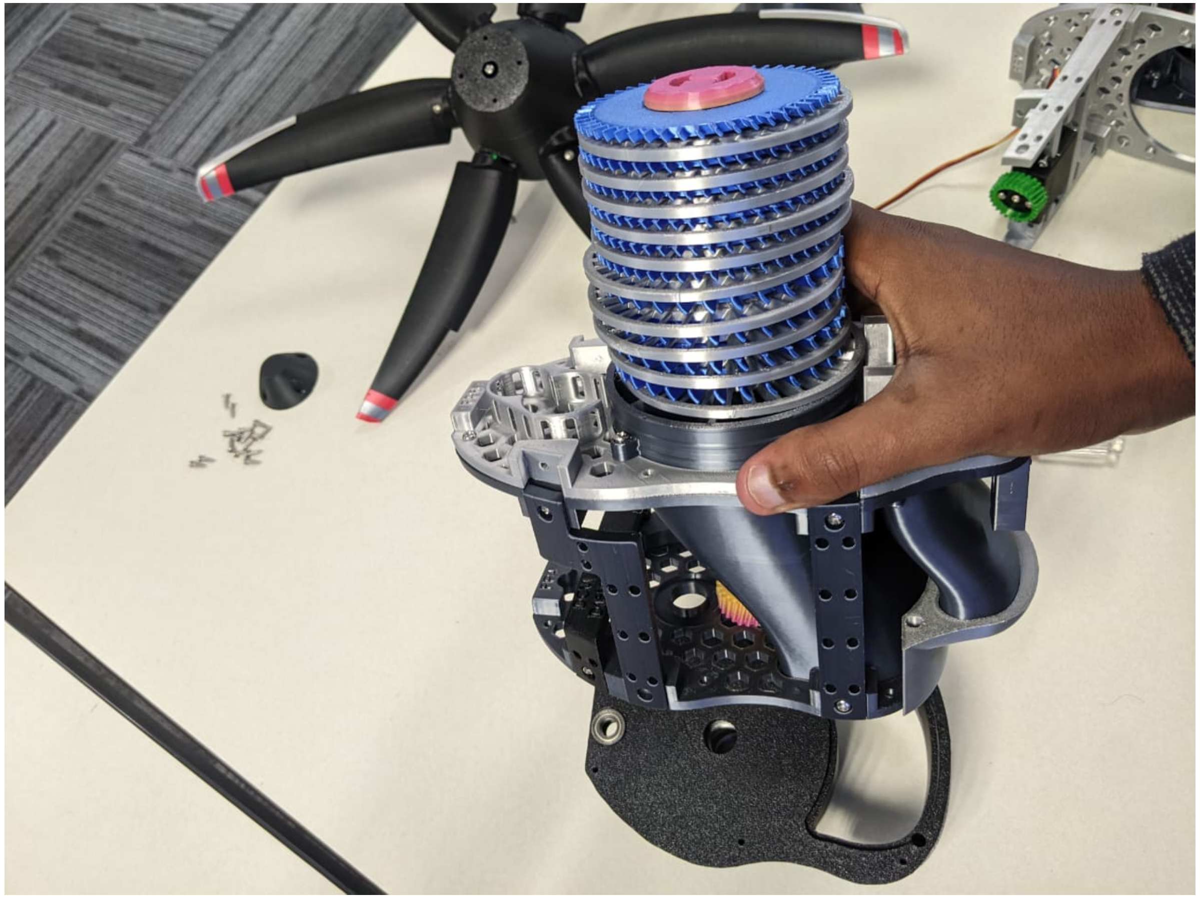

Image 5 (5.jpeg)

Say: Position the cylindrical core body on the lower bracket.

Do: Align the blue and silver engine core with the lower black bracket/base.

Visible check: The bracket and core should be centered before screws are inserted.

STEP 05 Step 5 of 24

Prepare the propeller-side module

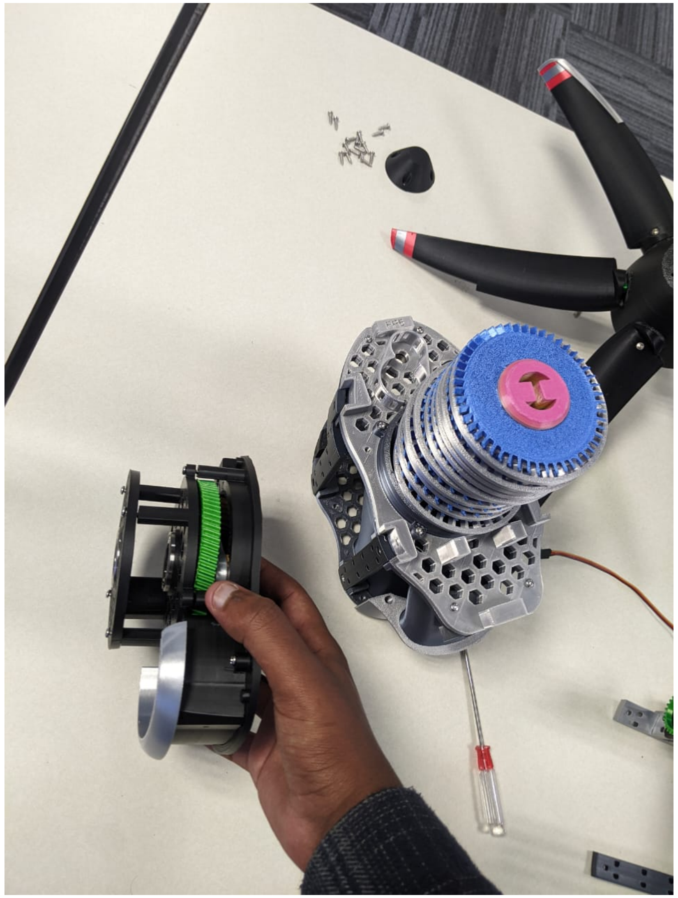

Image 6 (6.jpeg)

Say: Bring the propeller-side module toward the fan module.

Do: Hold the propeller-side module near the main engine body and show the mating side.

Visible check: Keep the green-ring side and body interface visible.

Connect lower and fan-side frame screws

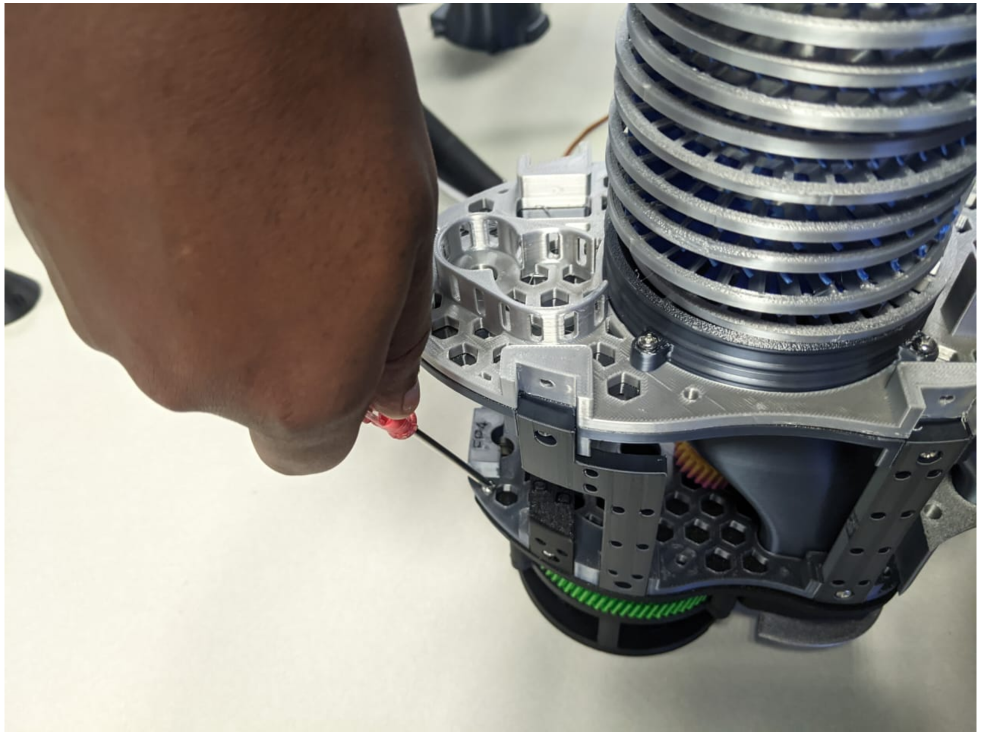

Image 7 (7.jpeg)

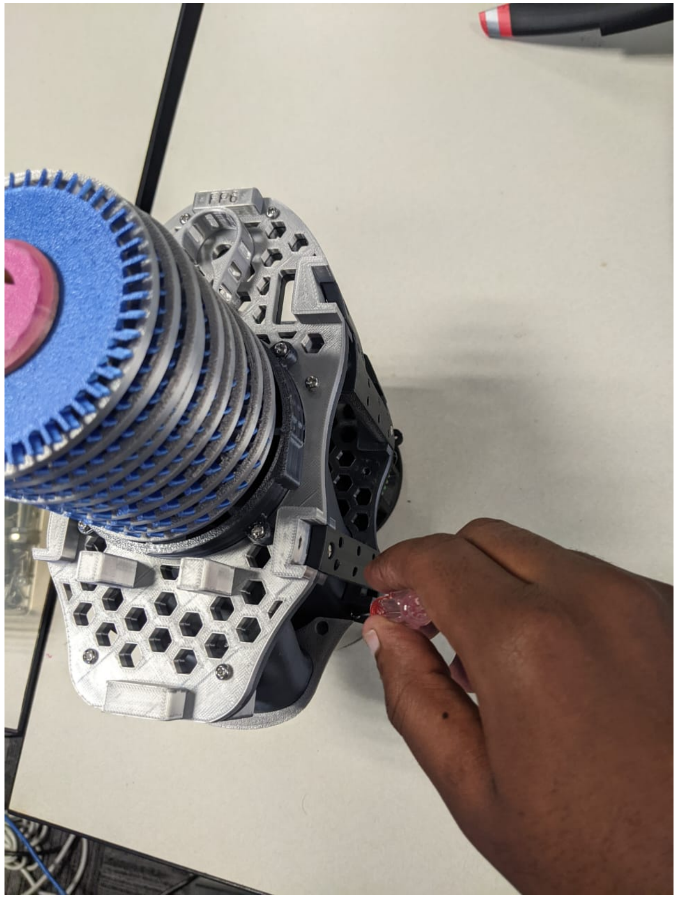

Image 8 (8.jpeg)

Say: Connect the lower side frame and fan-side frame using the screwdriver.

Do: Drive the screws through the lower side frame and fan-side mounting area.

Visible check: Use the full screw set for the interface, usually 2 or 4 screws. Do not move on after only one screw unless the part requires one.

STEP 07

Install the vertical rail frame

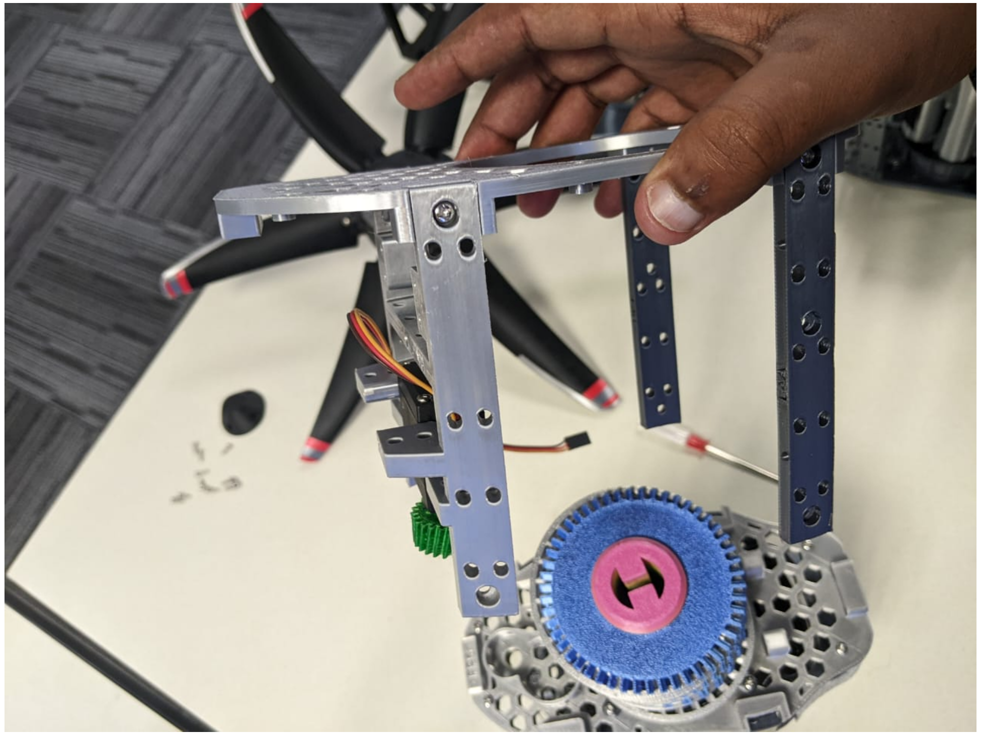

Image 9 (9.jpeg)

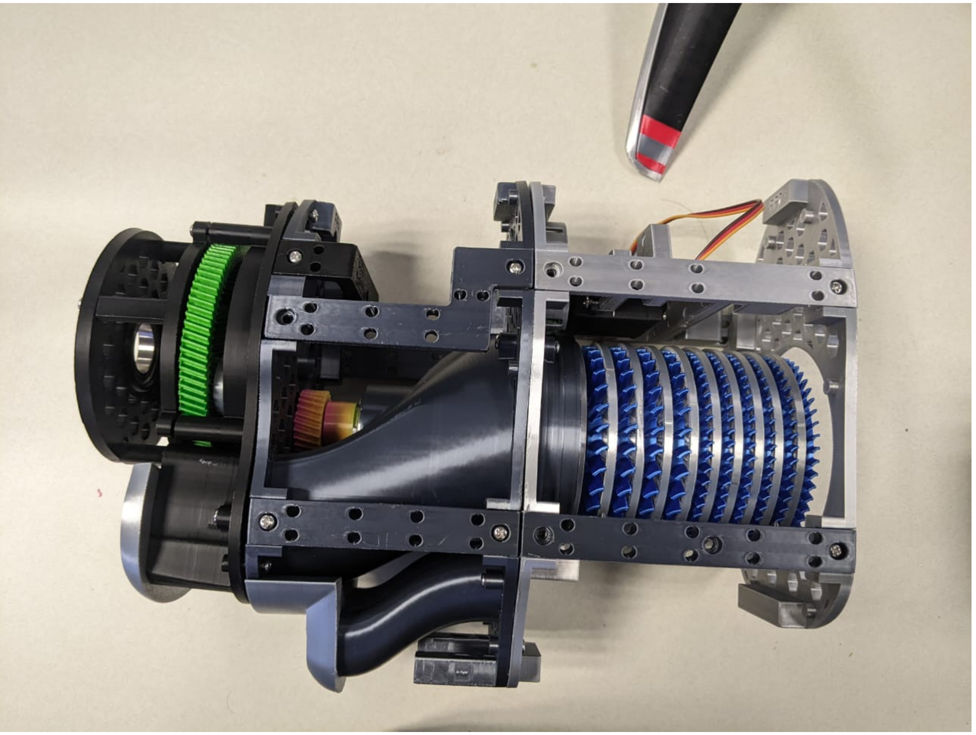

Image 10 (10.jpeg)

Say: Install the vertical rail frame and confirm alignment.

Do: Place the rectangular rail frame onto the fan/body module, then show the rail and body alignment.

Visible check: Rails should sit parallel to the cylindrical core and align with mounting holes.

STEP 08 Step 8 of 24

Secure the side rail and fan frame

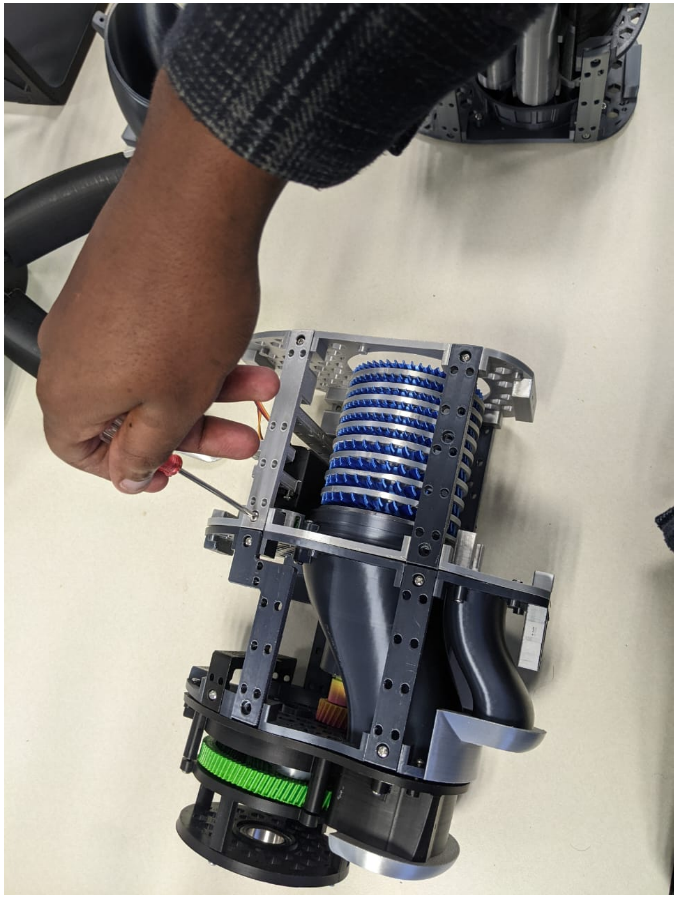

Image 11 (11.jpeg)

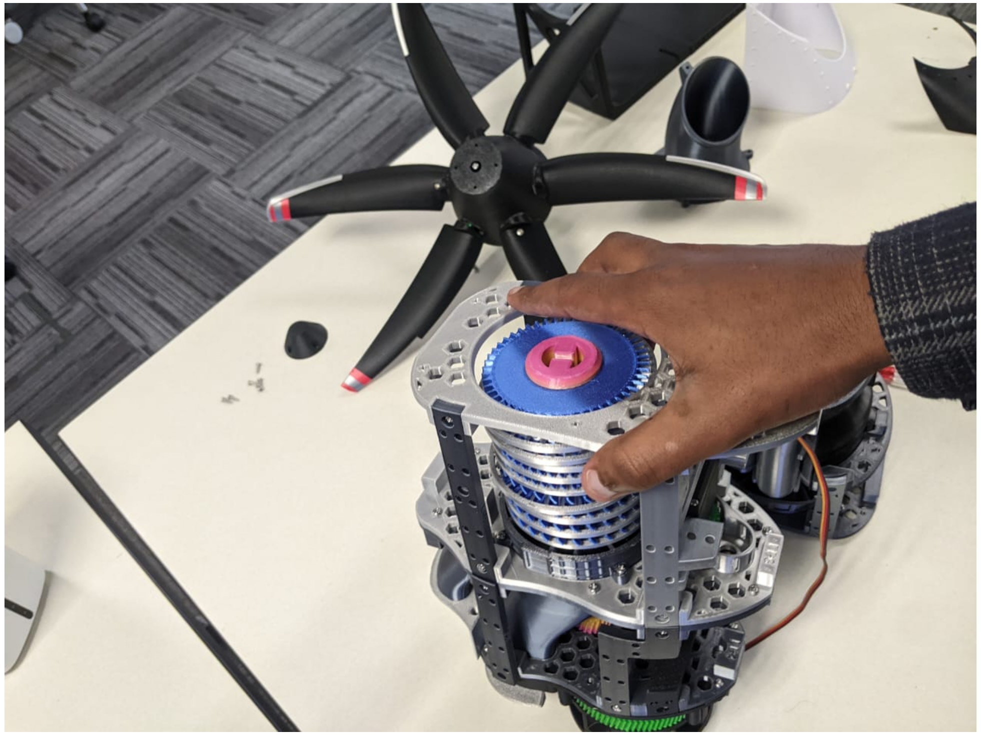

Image 12 (12.jpeg)

Say: Secure the side rail and fan frame screws.

Do: Use the screwdriver on the side rail and fan-frame fasteners while stabilizing the fan intake.

Visible check: Use the required 2 or 4 screws. The screwdriver tip and target screw must be visible.

STEP 09 Step 9 of 24

Present the body and fan modules

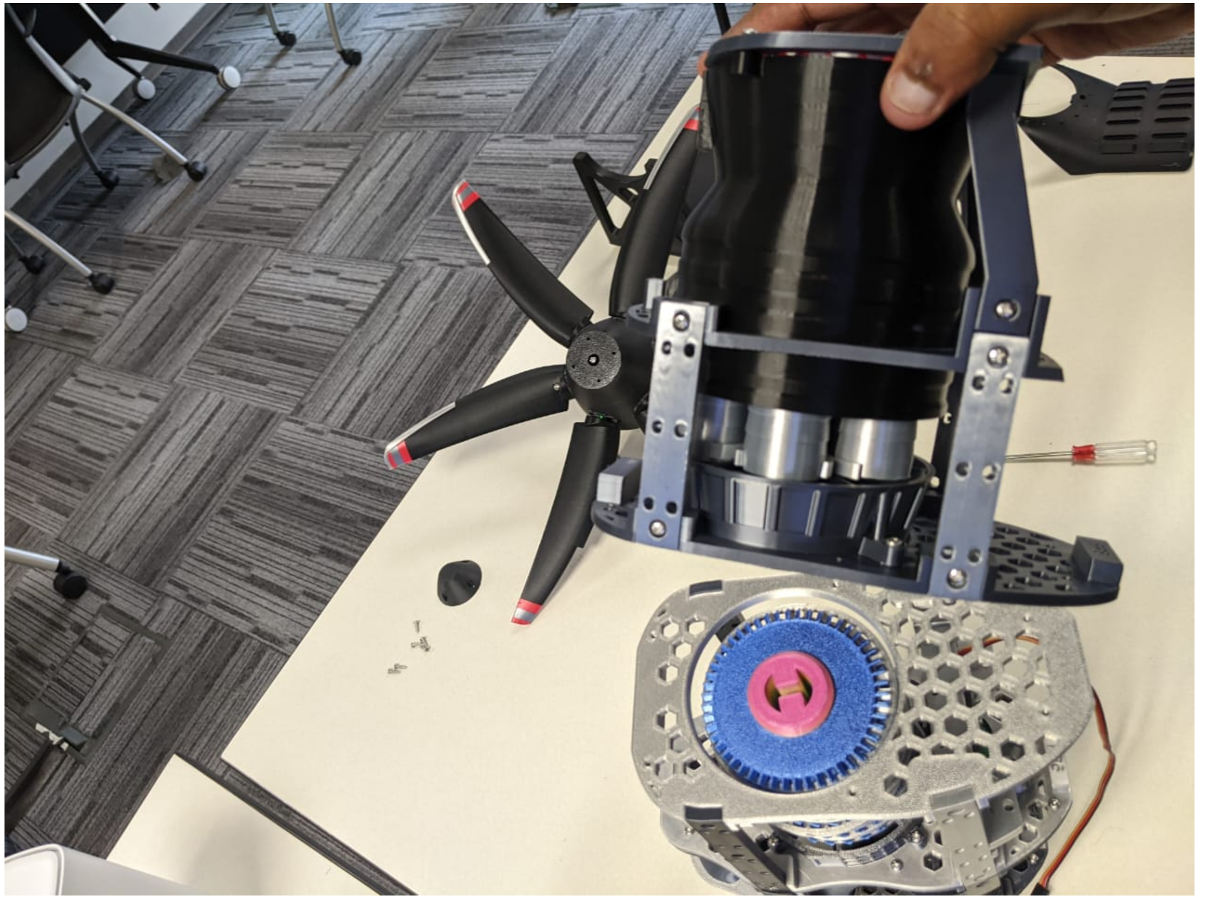

Image 13 (13.jpeg)

Say: Present the body and fan modules before closing the intake.

Do: Hold the body/frame module and fan module so the joining area is visible.

Visible check: Pause briefly so annotators can see the state before covers are installed.

Inspect the upright fan frame

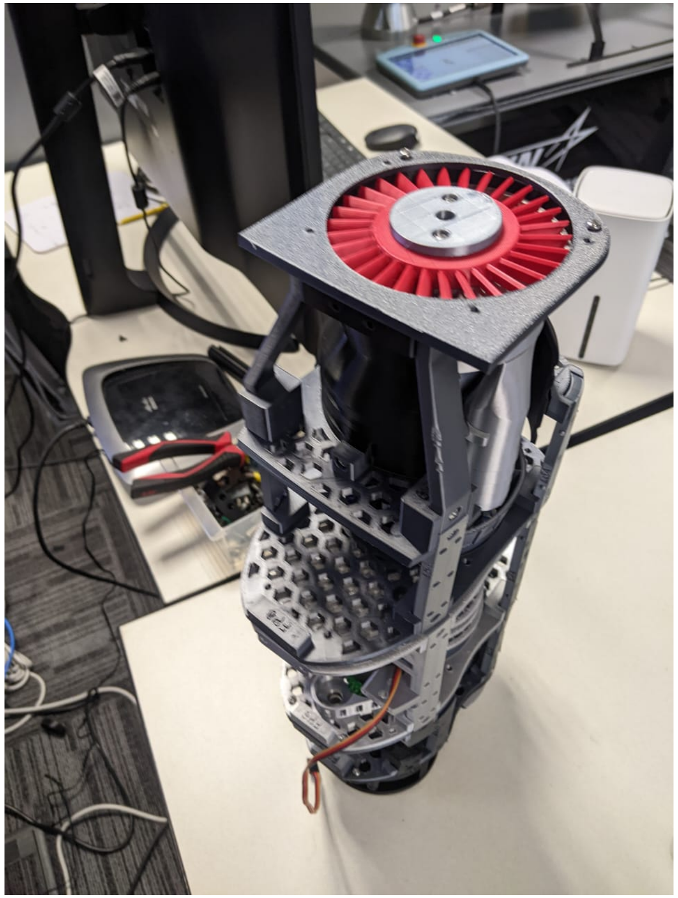

Image 14 (14.jpeg)

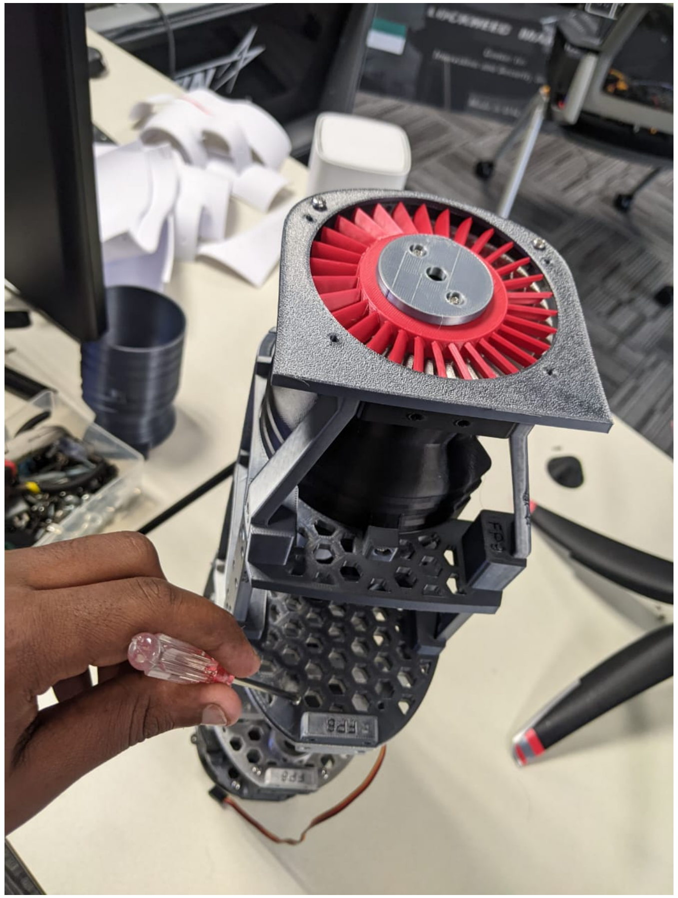

Image 15 (15.jpeg)

Say: Show the upright fan frame and exposed fan intake.

Do: Hold the assembly upright and show the red fan/intake, wiring, and frame rails.

Visible check: This is an inspection state. Do not cover the intake yet.

Lay the engine body horizontally

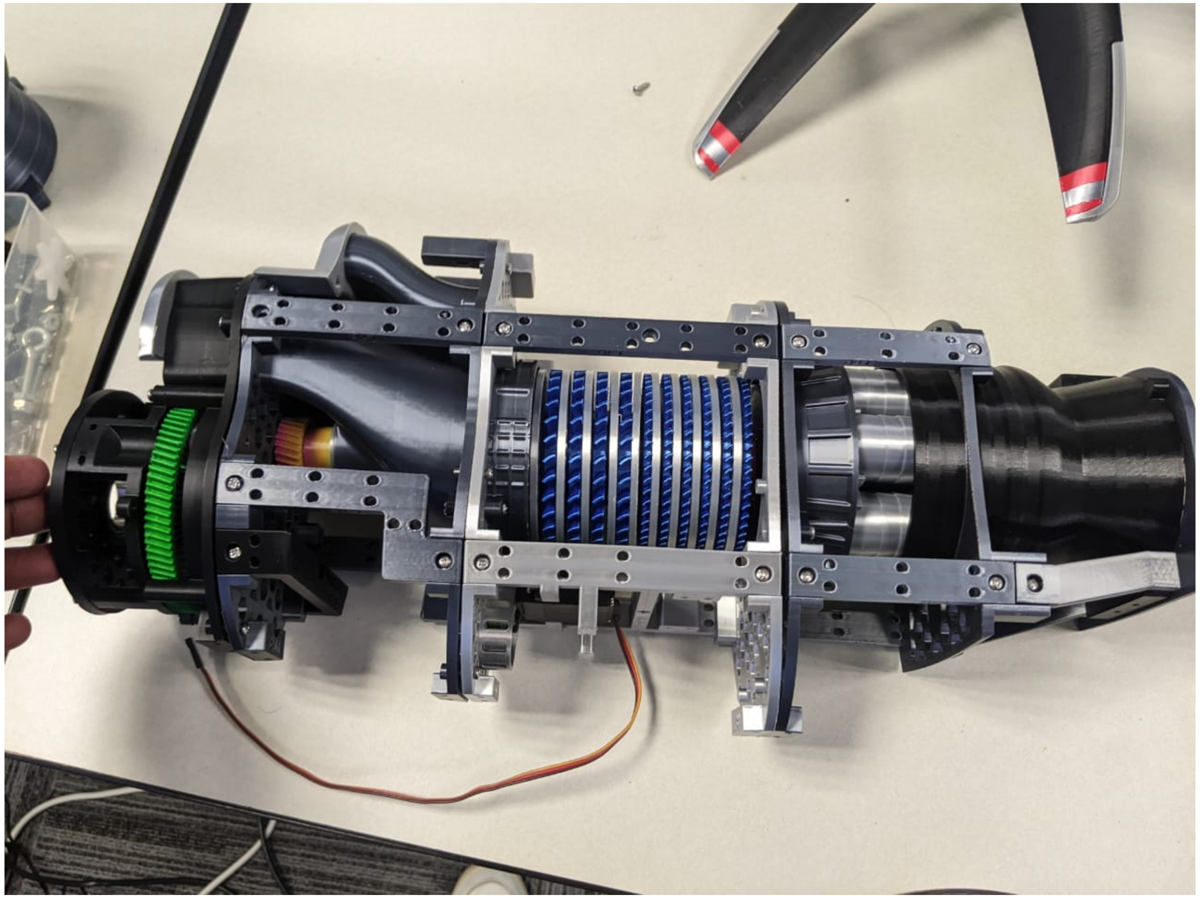

Image 16 (16.jpeg)

Say: Lay the cylindrical engine body horizontally.

Do: Place the main engine body flat on the table with the frame rails visible.

Visible check: The core, rails, and rear green ring should all be visible.

STEP 12 Step 12 of 24

Install the black nose cover

Image 17 (17.jpeg)

Image 18 (18.jpeg)

Say: Install the black nose cover over the fan opening.

Do: Align the black nose or duct cover over the fan/intake area and seat it on the upright assembly.

Visible check: The cover should sit flush and face the correct direction.

STEP 13

Align the white cowling

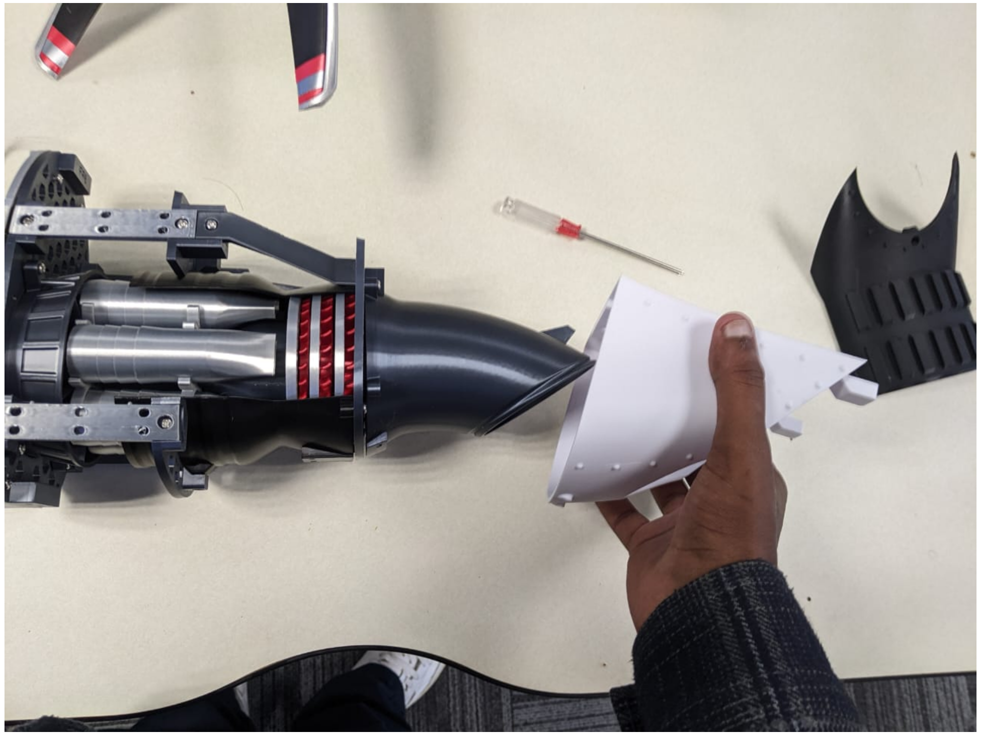

Image 19 (19.jpeg)

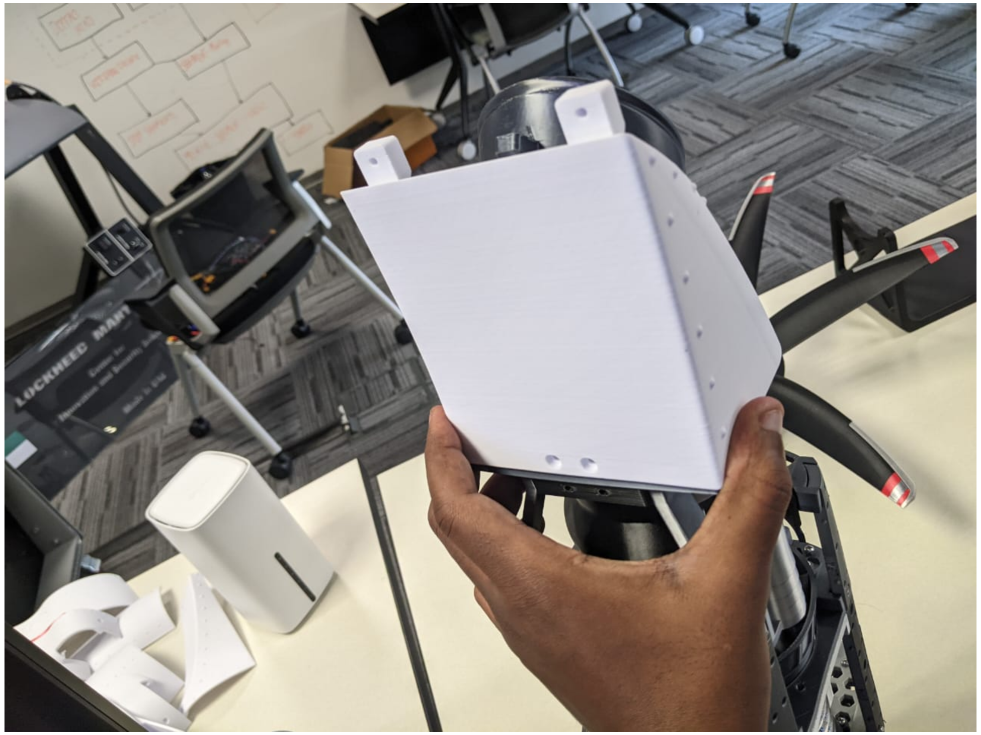

Image 20 (20.jpeg)

Say: Bring the white cowling to the nose cover and align it around the body.

Do: Hold the white cowling next to the black cover, then place it around the cylindrical body.

Visible check: The opening and curved surfaces should match the engine body orientation.

STEP 14

Secure the white cowling

Image 21 (21.jpeg)

Image 22 (22.jpeg)

Say: Secure the white cowling using the screwdriver.

Do: Drive the top cowling screws through the white cowling into the mounting points.

Visible check: Use the required 2 screws where applicable. Capture the screw head before and during fastening.

STEP 15 Step 15 of 24

Fit the black duct insert

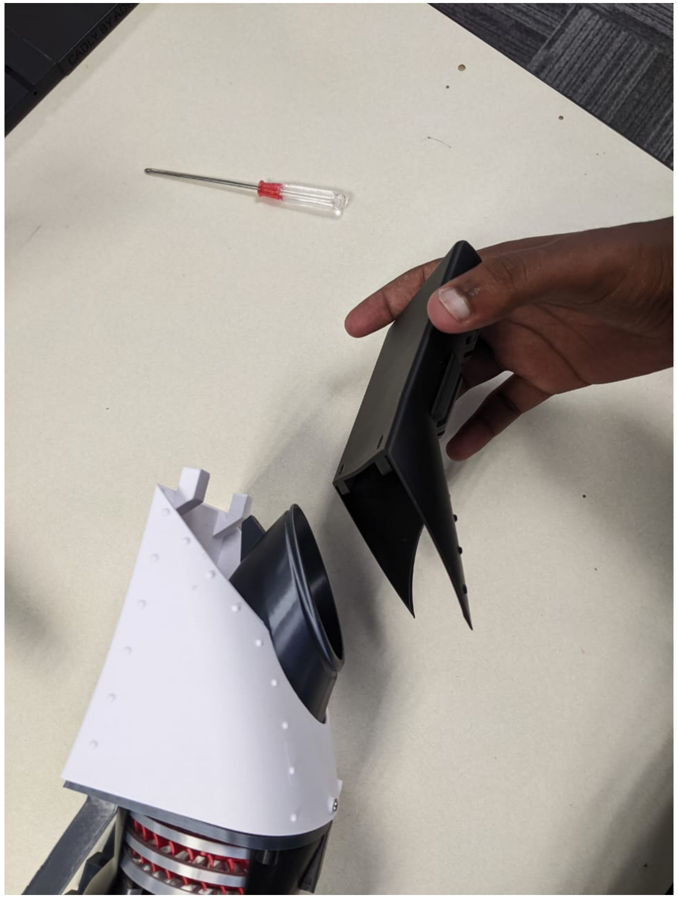

Image 24 (24.jpeg)

Say: Fit the black duct insert into the white cowling.

Do: Align the black insert with the cowling opening and press it into its seated position.

Visible check: The insert must be inside the cowling, not just held above it.

STEP 16 Step 16 of 24

Prepare the black duct cover

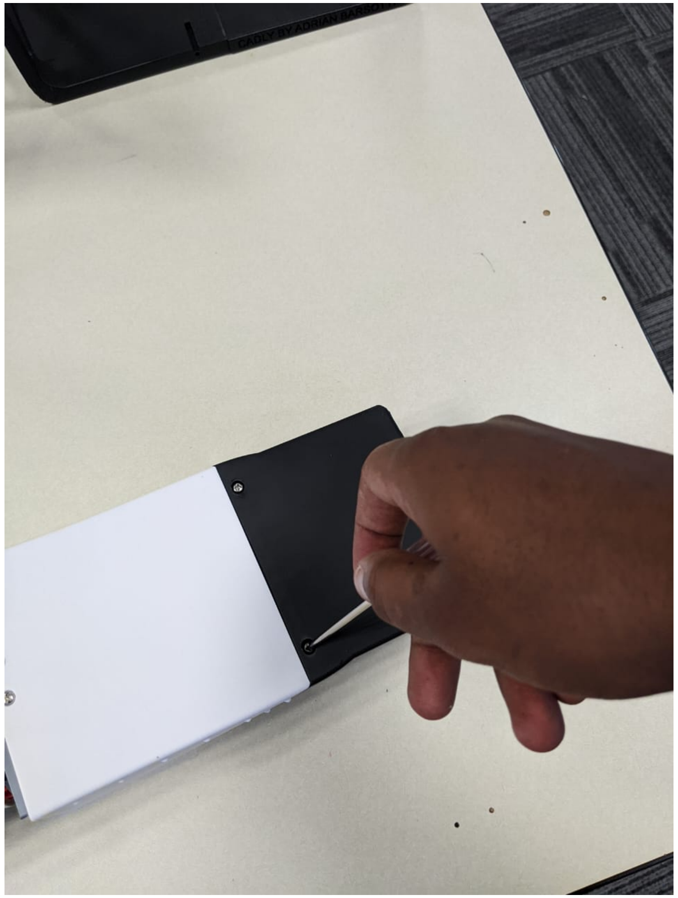

Image 25 (25.jpeg)

Say: Pick up the black duct cover.

Do: Lift the black duct cover and show its orientation before installing or placing it.

Visible check: Show the inner and outer surfaces if possible.

STEP 17 Step 17 of 24

Check the mounted cowling

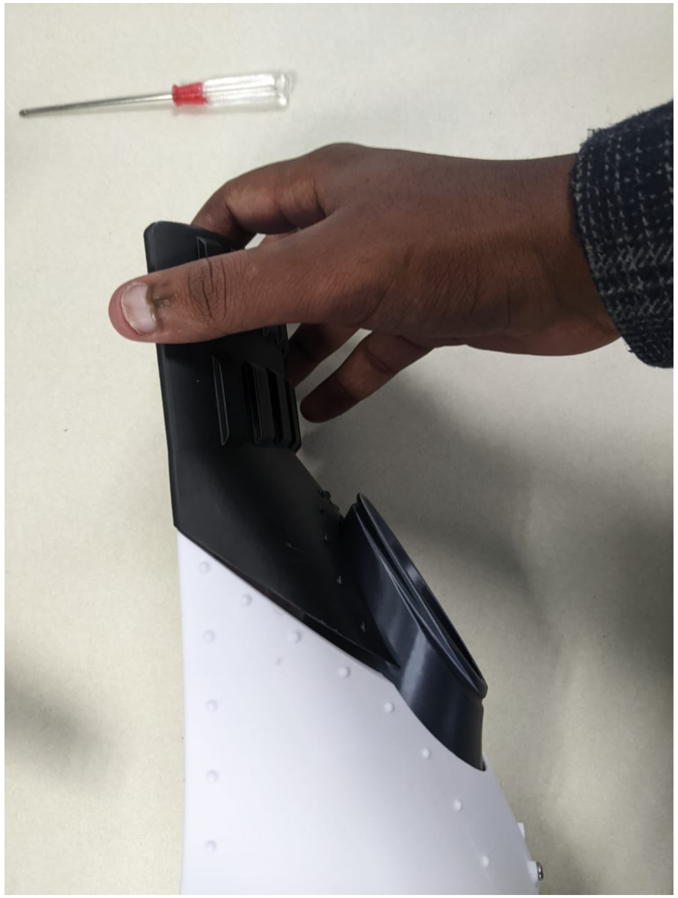

Image 26 (26.jpeg)

Say: Check the mounted cowling on the engine body.

Do: Touch or press the cowling/body area to confirm it is seated.

Visible check: The cowling should not visibly shift when checked.

Align the propeller-side module

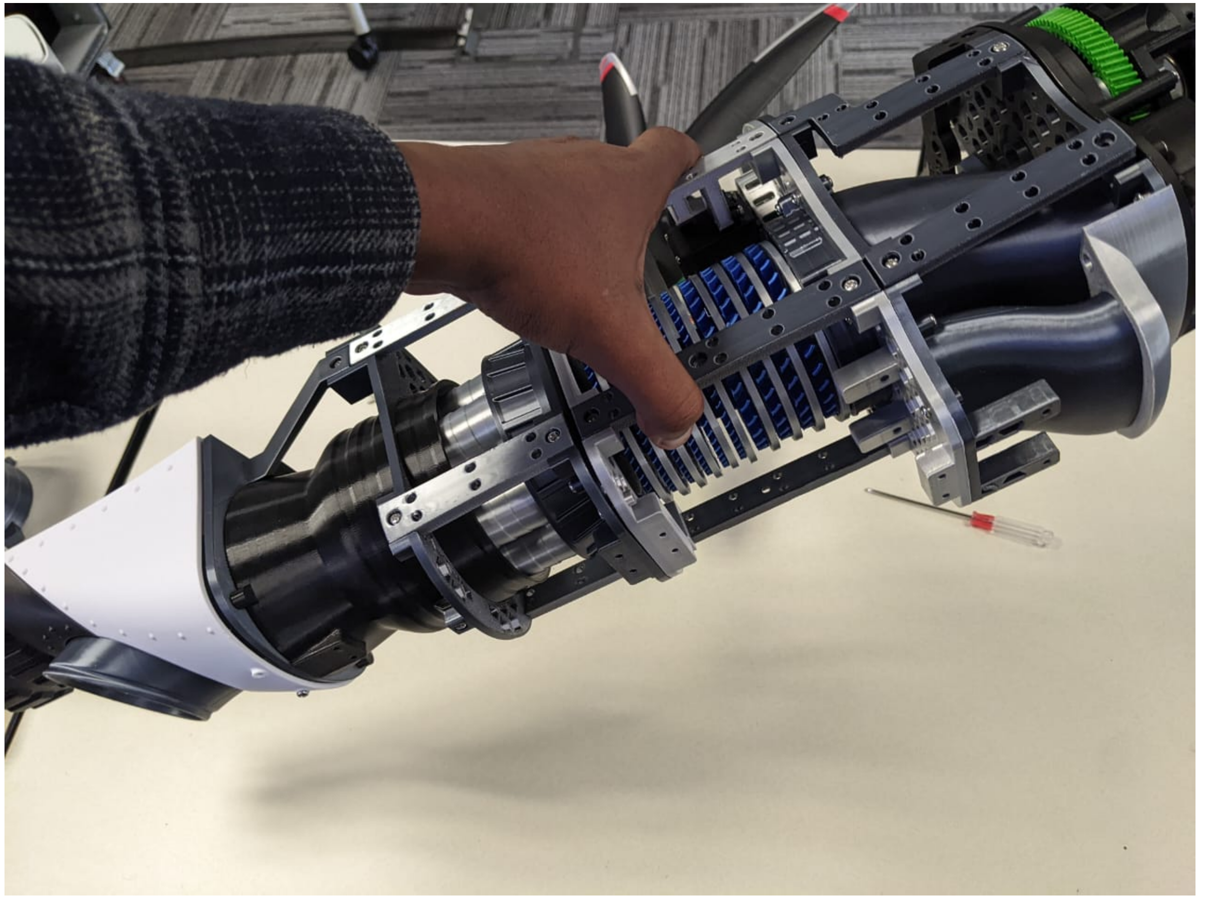

Image 27 (27.jpeg)

Say: Bring the propeller-side module into position.

Do: Move the propeller-side module toward the main body while keeping the mating parts visible.

Visible check: The module should be aligned before screw fastening begins.

STEP 19 Step 19 of 24

Connect the propeller-side module

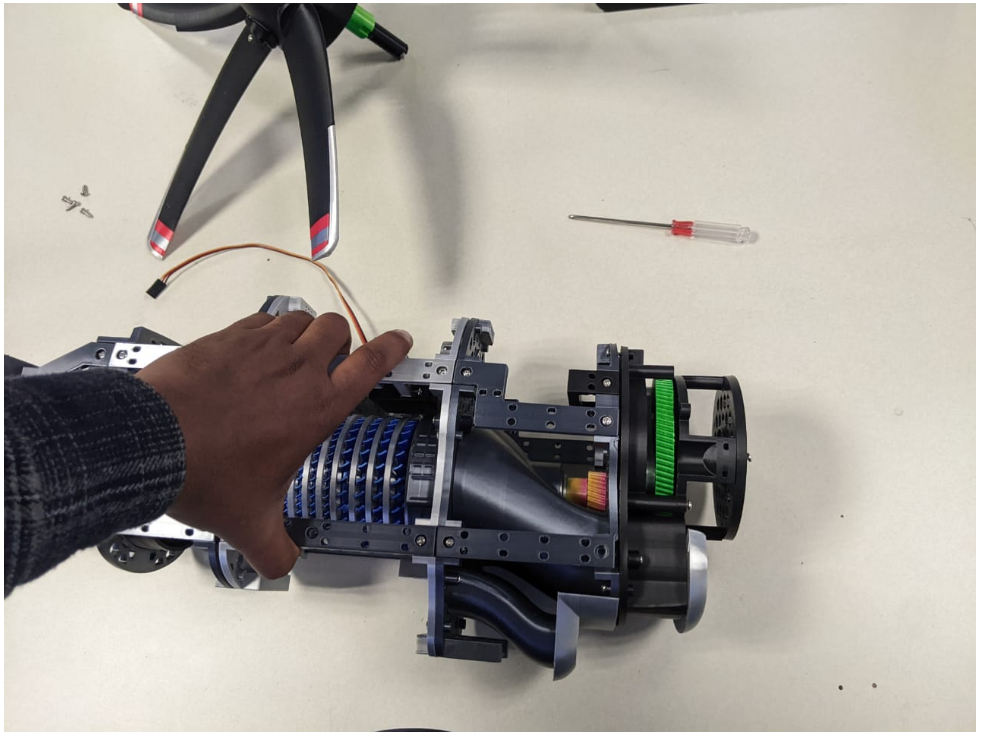

Image 28 (28.jpeg)

Image 29 (29.jpeg)

Say: Connect the propeller-side module to the engine body.

Do: Seat the propeller-side module against the cylindrical body and frame.

Visible check: The green ring and propeller-side body should align with the main frame.

STEP 20 Step 20 of 24

Stage the small propeller screws

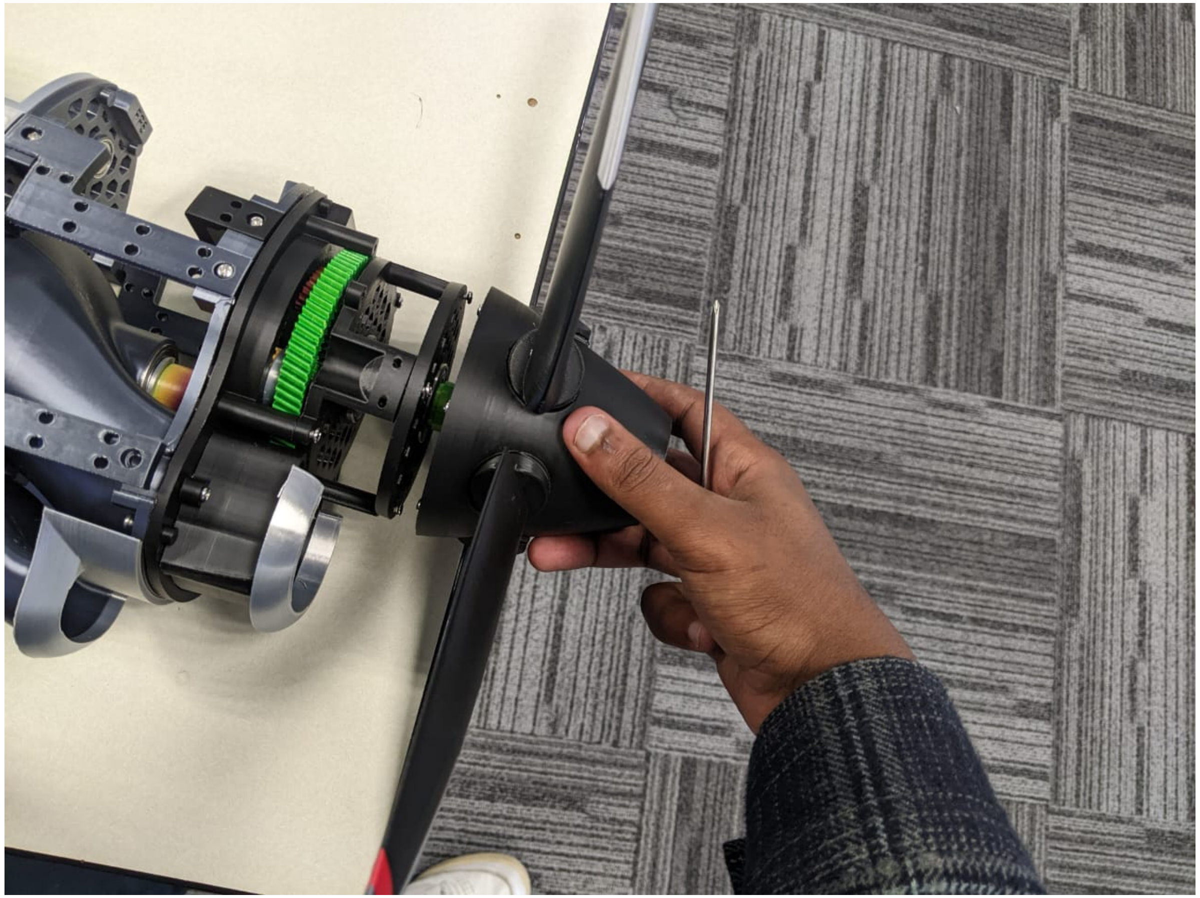

Image 30 (30.jpeg)

Say: Prepare the small screws for the propeller connection.

Do: Show the small screws on the table before using them on the propeller hub.

Visible check: These are the small screws for the propeller hub. Do not mix them with larger frame screws.

Secure the propeller hub with small screws

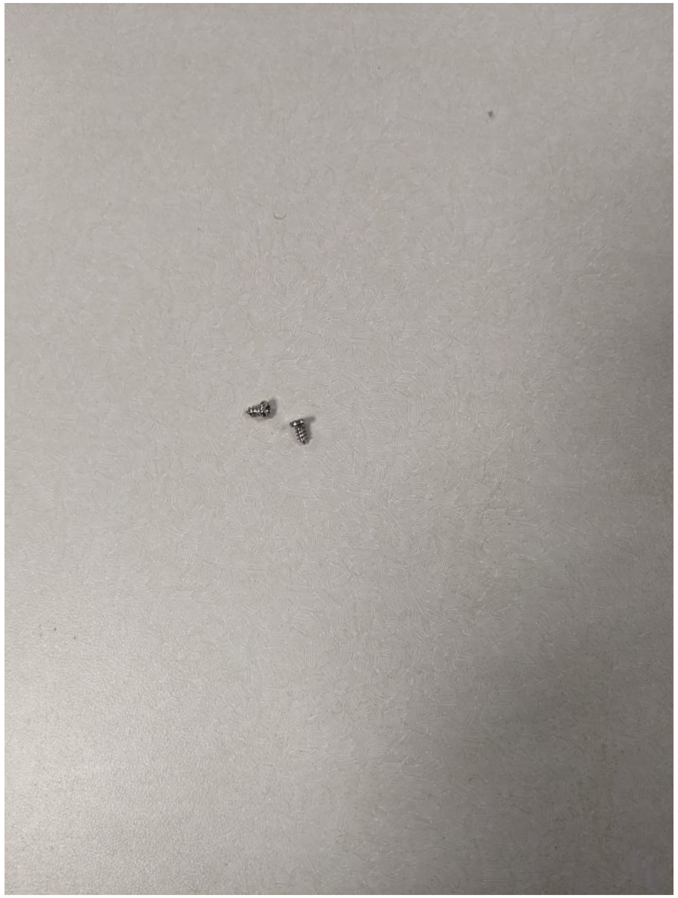

Image 31 (31.jpeg)

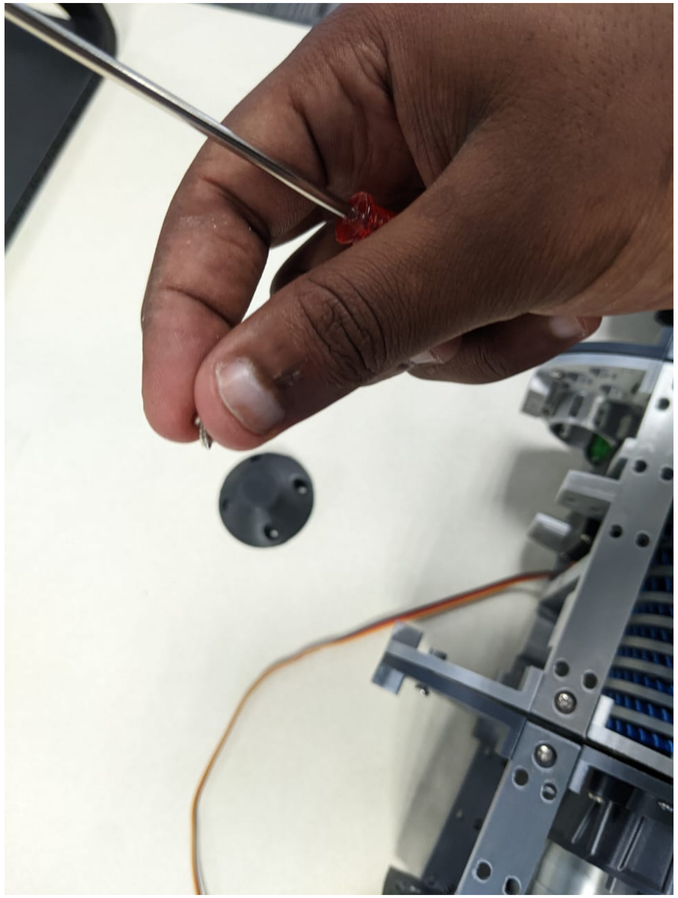

Image 32 (32.jpeg)

Say: Secure the propeller hub using the small screws.

Do: Position the small hub cap or fastener, then use the screwdriver to fasten the propeller hub.

Visible check: Use the small screws staged in the previous step. Capture each screw placement if 2, 4, or 6 screws are used.

STEP 22 Step 22 of 24

Prepare the black support frame

Image 33 (33.jpeg)

Say: Prepare the black support frame.

Do: Pick up or position the black support or duct frame beside the engine assembly.

Visible check: Show the frame orientation before connecting it.

STEP 23 Step 23 of 24

Check propeller-side rails and frame

Image 34 (34.jpeg)

Say: Check the propeller-side rails and main engine frame.

Do: Touch or steady the rails and main body to confirm the frame is aligned.

Visible check: This is a final alignment check before the end-state view.

STEP 24

Show the assembled engine model

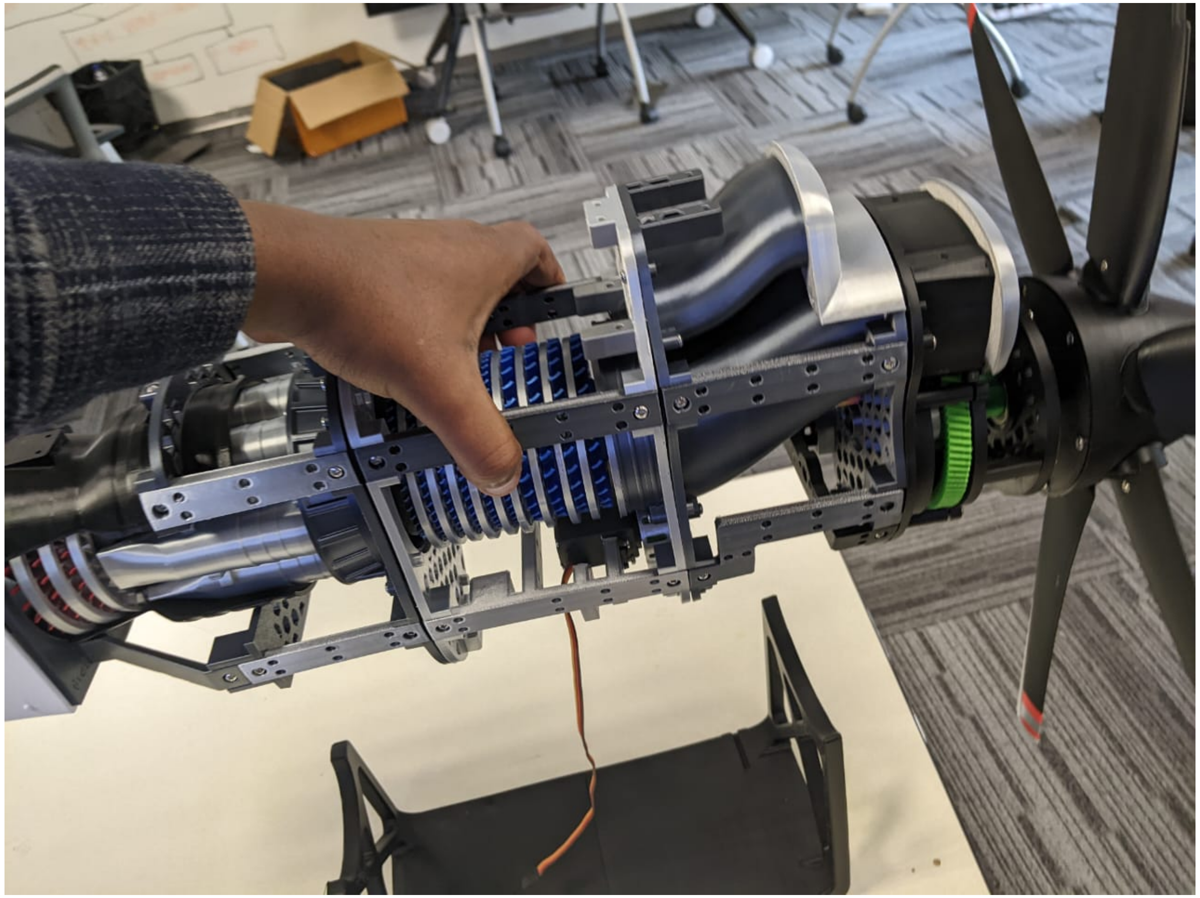

Image 35 (35.jpeg)

Say: Show the assembled engine model.

Do: Hold the final assembled state in view with the propeller, body, cowling, and frame visible.

Visible check: Pause for at least five seconds before stopping the recording.

Collector Quality Checklist

Visibility	Part, screw, screwdriver tip, and hands remain visible during the action.
Screw completeness	All required screws for the interface are used: 2, 4, or 6 depending on the part.
Propeller screws	Small screws from ordered Image 30 are used for the propeller hub.
State pause	The collector pauses after each completed state.
Mistake handling	Natural mistakes are kept in the recording and recovered without restarting.
Final view	The finished engine model is held still for at least five seconds.

Recommended file name

participant_session_trial_variant_hololens.mp4